The CFAC Theorem:

Doi–Peliti Generating Functions Are Lagrange Equations

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Abstract

For any chemical reaction network with polynomial vertex structure, the Doi–Peliti dressed-propagator generating function satisfies a Lagrange equation in the sense of Flajolet–Sedgewick analytic combinatorics. The recognition is an elementary rewriting; the consequences are the transfer theorem for all-orders coefficient asymptotics, classification of mean-field universality by Puiseux exponent, and a branch-gap contour integral for non-perturbative nucleation rates. We call the resulting framework Coupled Field Analytic Combinatorics (CFAC). This note states the theorem, gives the proof, and lists what can be read off from any reaction network.

1 Definitions

Definition 1 (Chemical reaction network)**.** A chemical reaction network (CRN) is a finite collection of species $\{A_1, \ldots, A_s\}$ and reactions $\left\{\sum_i k_{ri}\, A_i \to \sum_i l_{ri}\, A_i\right\}_r$ with nonnegative integer stoichiometry $k_{ri}, l_{ri} \in \mathbb{Z}_{\geq 0}$ and rates $k_r > 0$.

Definition 2 (Liouvillian of a CRN)**.** Associated to each reaction r are Heisenberg–Weyl creation and annihilation variables $\tilde{\psi}_i, \psi_i$ for each species A_i, and an operator

$$\widehat{Q}_r = k_r \Big[\prod_i \tilde{\psi}_i^{\,l_{ri}} - \prod_i \tilde{\psi}_i^{\,k_{ri}}\Big] \prod_i \psi_i^{k_{ri}} \tag{1}$$

in the "ordinary" (non-Doi-shifted) form, or equivalently

$$Q_r = k_r \Big[\prod_i (1+\tilde{\psi}_i)^{l_{ri}} - \prod_i (1+\tilde{\psi}_i)^{k_{ri}}\Big] \prod_i \psi_i^{k_{ri}} \tag{2}$$

in the Doi-shifted form. The Liouvillian of the CRN is $Q = \sum_r Q_r$ (or $\widehat{Q} = \sum_r \widehat{Q}_r$).

The two forms (1)–(2) are related by the Doi shift $\tilde{\psi}_i \to 1 + \tilde{\psi}_i$ (Doi, 1976; Peliti, 1985; Täuber, 2014). Expanding any product $\prod_i (1+\tilde{\psi}_i)^{l_{ri}}$ or $\prod_i \tilde{\psi}_i^{l_{ri}}$ and collecting like terms gives a *vertex dictionary*:

$$Q \;=\; \sum_{(m_1+n_1\geq 1,\ldots)} g_{\{m\},\{n\}} \prod_i \tilde{\psi}_i^{m_i}\, \psi_i^{n_i}, \tag{3}$$

with coupling coefficients $g_{\{m\},\{n\}}$ determined algorithmically by the stoichiometry and rates (Amarteifio, 2019). Each term $\prod_i \tilde{\psi}_i^{m_i} \psi_i^{n_i}$ is an interaction vertex; its n_∂ value is zero unless spatial gradients are added by hand (e.g. to model drift or chemotaxis).

Definition 3 (Polynomial vertex structure)**.** A CRN has *polynomial vertex structure* if the Liouvillian, after the Doi shift (2) and possibly the projection to zero spatial dimensions (replacing $\psi_i(\mathbf{x}, t) \to \psi_i(t)$), expands into a finite sum of monomials $g_{\{m\},\{n\}} \prod_i \tilde{\psi}_i^{m_i} \psi_i^{n_i}$ with no momentum-dependent couplings.

Every CRN with finite stoichiometry has polynomial vertex structure in this sense. Continuous gradient vertices (e.g. the chemotaxis term $\chi\,\tilde{\psi}(\nabla\phi)(\nabla\psi)$) are *not* of this form at nonzero momentum; their zero-momentum / well-mixed projections are.

Definition 4 (Dressed-propagator generating function)**.** Fix a species A with self-coupling vertices only (multi-species cases are handled by resultants; see §7). The dressed propagator is $G(z) = \sum_{n\geq 0} G_n z^n$, where G_n is the sum over connected 1PI diagrams contributing to the two-point function at order n in the bare propagator z. The *vertex function* is

$$\phi(G) \;=\; 1 + \sum_{m+n\geq 3} g_{mn}\, G^{m+n-2}, \tag{4}$$

summing over all vertices $g_{mn}\tilde{\psi}^m\psi^n$ of the Liouvillian with $m+n \geq 3$. (Two-leg vertices $m+n = 2$ are mass terms absorbed into z.)

2 The CFAC theorem

Theorem 1 (CFAC recognition)**.** *For any CRN with polynomial vertex structure, the dressed-propagator generating function $G(z)$ and the vertex function $\phi(G)$ of Eq. (4) satisfy*

$$\boxed{G(z) \;=\; z \cdot \phi(G(z))} \tag{5}$$

as a closed functional equation over formal power series in z. Equation (5) is a Lagrange equation *in the sense of analytic combinatorics (Flajolet and Sedgewick, 2009).*

Consequently, for any such CRN the following quantities are determined directly by the polynomial ϕ:

(C1) All-orders coefficient asymptotics. *If z_* is the radius of convergence of $G(z)$, with Puiseux exponent p/q at z_*, then by the transfer theorem*

$$[z^n]\, G(z) \;\sim\; \frac{C}{\Gamma(-p/q)}\, n^{-p/q-1}\, z_*^{-n}, \qquad n \to \infty, \tag{6}$$

with C set by the local behaviour of ϕ near (G_*, z_*).

(C2) Mean-field universality class. *The rational Puiseux exponent p/q at the dominant branch point of (5) classifies the mean-field scaling: $1/2$ is the directed-percolation / pair-annihilation class, $1/3$ is the compact-DP / triple-root class, higher-order mergers give further multi-critical classes.*

(C3) Non-perturbative nucleation barrier. *When ϕ produces a bistable effective dynamics, the nucleation barrier is the contour integral*

$$S \;=\; \int_1^{z_*} \frac{G_{\text{unst}}(z) - G_{\text{stab}}(z)}{z}\, \mathrm{d}z, \tag{7}$$

equal to the Freidlin–Wentzell / WKB quasi-potential $\int_{x_1}^{x_2} \ln[w^-/w^+]\,\mathrm{d}y$ by change of variables.

(C4) UV divergence structure. *At spatial dimension d, the superficial degree of divergence of a 1PI graph with L loops, I internal lines, and n_∂^{tot} gradient insertions is $\omega = dL - 2I + n_\partial^{\text{tot}}$. Power counting on the finite enumeration of 1PI graphs consistent with the vertex dictionary is then purely algebraic.*

Scope. The theorem as stated holds when ϕ is a polynomial in G alone. This covers every CRN with finite stoichiometry at zero spatial dimensions or after integrating out additional field variables at steady state. When gradient vertices carry momentum-dependent couplings, ϕ becomes a functional of momentum-dependent propagators; the recognition extends, but the transfer-theorem machinery becomes functional-analytic. All applications in this note stay within the polynomial scope.

Corollary 2 (CFAC pipeline)**.** *For any CRN with polynomial vertex structure, the following chain is purely algebraic and mechanical:*

stoichiometry $\to$ *Liouvillian* Q $\to$ *vertex dictionary* $\{g_{mn}\}$ $\to$ *vertex function* $\phi(G)$ $\to$ *Lagrange equation* $G = z\phi(G)$ $\to$ *(C1) asymptotics, (C2) universality, (C3) nucleation, (C4) divergences.*

No loop integral, no saddle-point of an action, no WKB trajectory is required to reach any of (C1)–(C4).

Corollary 3 (Rare-event scaling)**.** *For a bistable CRN with polynomial vertex structure and effective system size* V*, the mean first-passage time from the metastable state to the unstable barrier is*

$$\tau(V) \;=\; C\,\mathrm{e}^{VS}\,[1 + O(1/V)], \tag{8}$$

with barrier S *given by the branch-gap integral (7) of* $\phi(G)$ *and prefactor* C *determined by the local Puiseux data at the branch point via the transfer theorem (6). Neither direct stochastic simulation nor action saddle-pointing is required to extract* S*.*

Proof. Equation (8) is the large-deviation form of the Freidlin–Wentzell escape rate (Freidlin and Wentzell, 2012). By (C3), S equals the branch-gap integral of the Lagrange equation (5). The prefactor C comes from the Gaussian fluctuations around the instanton trajectory; in the Lagrange-equation language these fluctuations are controlled by the local coefficients of the Puiseux expansion at (G_*, z_*), which feed into (6). The two pieces of data (S, C) are thus both read off from the polynomial ϕ without simulation or WKB. (In practice, as in (Amarteifio, 2026b), C is often calibrated from small-V simulation; the in-principle route via (6) is the one guaranteed by Theorem 1.) □

Corollary 4 (Coupled field extension via resultant)**.** *For a CRN on* s *species with polynomial vertex structure across all species, the system of Dyson–Schwinger equations* $G_i = z_i\,\phi_i(\{G_j\})$ *for* $i = 1, \ldots, s$ *admits a univariate reduction: the resultant* $\mathcal{R}(G_1, z_1; \{z_j\}_{j\neq 1})$ *obtained by eliminating* $G_2, \ldots, G_s$ *from the system is a polynomial in* G_1 *whose roots* $G_1(z_1)$ *inherit a Lagrange-equation structure. Theorem 1 applies to the resultant with the following scope:*

- *Consequences (C1), (C2), (C4)—coefficient asymptotics, Puiseux classification of mean-field universality, and UV divergence structure—hold with* ϕ *replaced by* $\mathcal{R}$*, directly and unconditionally.*
- *Consequence (C3)—the branch-gap instanton integral—holds when the physical nucleation saddle is a real branch point of* $\mathcal{R}$*. It can fail when the saddle migrates into the complex analytic continuation of the resultant.*

Proof sketch and scope. The resultant of a polynomial system is itself polynomial in the retained variable (Cox et al., 2007); (C1), (C2), (C4) apply mechanically. For (C3), a numerical test on a 3-site coupled ring reported in (Amarteifio, 2026b) shows that the branch-gap formula on the resultant agrees with direct simulation in the decoupled limit (2.6%), under-predicts the barrier by ∼20% in a weakly coupled regime, and has no real branch point at all in a moderately coupled regime, even though the simulated nucleation rate remains exponential. The nucleation saddle has left the real algebraic variety. Establishing (C3) in that regime requires complex-analytic continuation of the resultant, outside the polynomial-vertex scope of Theorem 1. □

Corollary 5 (One-loop algebraic bridge via tropical geometry)**.** *For any one-loop two-propagator integral arising in a CRN with polynomial vertex structure, the bridge integral (C4) admits a* closed algebraic form*: the Arkani-Hamed–Hillman–Salvatori leading-facet-plus-Hessian evaluation of the Schwinger representation is* exact*, not asymptotic. Explicitly, with first Symanzik polynomial* $F = m_1^2 u_1^2 + (p^2 + m_1^2 + m_2^2)u_1 u_2 + m_2^2 u_2^2$,

$$I_{\mathrm{DP}}(p; m_1, m_2) \;=\; \frac{1}{\sqrt{\lambda_E}}\,\ln\left[\frac{\left(\sqrt{\lambda_E} + p^2 + m_1^2 + m_2^2\right)^2}{4\,m_1^2\,m_2^2}\right], \tag{9}$$

with $\lambda_E = (p^2{+}m_1^2{-}m_2^2)^2{+}4p^2m_2^2$ the Euclidean Källén discriminant (equivalently, the determinant of the tropical Hessian at the interior saddle). The branch cut $\lambda_E = 0$ locates the Minkowski threshold $s = (m_1{+}m_2)^2$ directly on the Newton polytope.

Proof sketch. F is homogeneous of degree $L{+}1 = 2$ in the Schwinger parameters; on the projective simplex it is a non-degenerate quadratic form with constant Hessian. The Schwinger integral of $F^{-d/2}$ is Gaussian against the leading-facet saddle, evaluated exactly by the saddle-plus-Hessian prescription (Arkani-Hamed et al., 2022; Borinsky, 2020): no higher Taylor corrections arise because $\ln F$ terminates at second order around the saddle up to a total derivative. Details, numerical verification across 14 orders of magnitude in p^2/m^2 and m_1^2/m_2^2, and the extension to the coupled DP–MSR bridge appear in (Amarteifio, 2026b). □ □

Corollary 5 sharpens (C4): at one loop, the bridge step of the CFAC pipeline (Corollary 2) is not a numerical dictionary but a closed algebraic expression. Combined with the recognition theorem, the consequence is that for any CRN with polynomial vertex structure, the entire one-loop beta function is determined by symbolic manipulation of two polynomials: $\phi(G)$ (from the vertex dictionary) and F (the first Symanzik of the relevant topology). Higher loops require genuine multi-facet sums over Newton polytopes, for which Borinsky's tropical Monte Carlo (Borinsky, 2020; Borinsky et al., 2025) gives an exact integration procedure.

Why one loop is often enough. The restriction of Corollary 5 to one loop is less severe than it appears, because (C4) itself is typically *one-loop-exact* for CRNs of interest. A CRN is one-loop exact when the -enumeration gives $\omega(L) < 0$ for every topology with $L \geq 2$—meaning no higher-loop UV divergences arise. This is the situation for pair annihilation, pure DP, BARW, chemotaxis (**?**), the KS Doi–Peliti/MSR coupling sector (**?**Amarteifio, 2026b), and more generally for any CRN whose vertex content is sufficiently constrained by finite stoichiometry that the enumeration collapses. In those cases Corollary 5 and (C4) combine to close the *entire* perturbative pipeline in symbolic form: reaction network → vertex dictionary → Lagrange equation → Puiseux exponent + branch-gap + closed-form bridge integral, with no numerical quadrature anywhere. The restricted class where this fails (e.g. active-birth ψ^4 sectors with persistent two-loop divergences) is exactly the class where -expansion resummation is also required in the standard direct-RG route; CFAC neither helps nor hurts there, and Borinsky's algorithm remains the available exact evaluator.

What these corollaries are (and are not). Corollary 3 is a statement about the large-V form of the MFPT, using the standard large-deviation structure (Freidlin and Wentzell, 2012). The novelty is that *both* the exponent S and the prefactor C are read off from ϕ; the present paper gives S rigorously and notes that C is in-principle determined by the transfer theorem. Corollary 4 is the technical observation that polynomial vertex structure across multiple species admits a univariate reduction by resultant, to which Theorem 1 applies unconditionally for (C1), (C2), (C4). For the nucleation consequence (C3), the reduction holds only when the physical saddle is a real branch point of the resultant; a 3-site lattice test in (Amarteifio, 2026b) finds this fails at moderate coupling, where the saddle is complex. Extending (C3) to that regime is an open problem and requires machinery beyond the polynomial scope of Theorem 1.

3 Proof

Proof of Theorem 1. The Dyson–Schwinger equation for the dressed propagator of a field theory with vertex content $\{g_{mn}\tilde{\psi}^m\psi^n\}$ is obtained by summing 1PI insertions into the bare propagator: $G = z{+}z\,\Sigma(G)\,G$, where the self-energy Σ is a sum over 1PI vertex insertions. A 1PI insertion of the vertex $g_{mn}\tilde{\psi}^m\psi^n$ contracts $(m{-}1)$ of its $\tilde{\psi}$-legs and n of its ψ-legs to internal dressed propagators, contributing $g_{mn}\,G^{m+n-2}$. Summing these plus the identity gives (4). Substituting into the DSE:

$$G \;=\; z + z\,\Sigma(G)\,G \;=\; z\big[1 + \Sigma(G)\,G\big] \;=\; z\,\phi(G),$$

which is (5). The identification of this as a Lagrange equation over formal power series is definitional (Flajolet and Sedgewick, 2009, Ch. VII). Consequences (C1)–(C4) then follow directly: (C1) is the transfer theorem; (C2) is the Newton-polygon analysis of (5) at the discriminant zero; (C3) is proved by change of variables $y \to z = w^-(y)/w^+(y)$ in the quasi-potential integral; (C4) is standard QFT power counting applied to the 1PI graph list produced by the algebraic formalism (Amarteifio, 2019). □

The proof of (C3) is worth spelling out explicitly because the form (7) appears to be new. Let $w^{\pm}(y)$ denote the effective forward and backward rates for the order-parameter y (concentration of A, obtained by taking the mean-field reduction of the CRN). The WKB quasi-potential (Freidlin and Wentzell, 2012; Elgart and Kamenev, 2004; Assaf and Meerson, 2017) is $\varphi(y) = \int_{x_1}^{y} \ln[w^-/w^+]\,\mathrm{d}y'$, and the nucleation barrier is $S = \varphi(x_2)$ between metastable state x_1 and unstable barrier x_2 (both satisfying $w^+ = w^-$). Substituting $z(y) = w^-(y)/w^+(y)$ gives $\mathrm{d}\varphi = (\ln z)\,(\mathrm{d}y/\mathrm{d}z)\,\mathrm{d}z$. Both endpoints x_1, x_2 map to $z = 1$ (fixed points), but the integration path switches from the stable branch of $y(z)$ to the unstable branch at the branch point z_*. Integration by parts, $\int (\ln z)\, y'(z)\,\mathrm{d}z = [y(z)\ln z] - \int y(z)/z\,\mathrm{d}z$, kills the boundary terms ($\ln 1 = 0$; branch values agree at z_*), leaving (7). □

4 Background

Two bodies of work make the recognition trivial to state.

The algebraic formalism for CRNs (Amarteifio, 2019) maps any chemical reaction network to the Liouvillian (2) (or (1)) algorithmically via the stoichiometry matrix, building on Doi's second-quantised representation (Doi, 1976), Peliti's coherent-state path integral (Peliti, 1985), and Täuber's systematic treatment (Täuber, 2014). The key algorithmic step is that the expansion of $\prod_i (1 + \tilde{\psi}_i)^{l_{ri}}$ into monomials is closed-form (multinomial expansion), so the vertex dictionary $\{g_{mn}\}$ is determined exactly by the stoichiometry and rates. No field-theoretic insight is required once the CRN is written down; the construction is mechanical.

Analytic combinatorics (Flajolet and Sedgewick, 2009) is the asymptotic theory of generating functions developed for enumerative combinatorics. Its central machinery is: Lagrange equations $G = z\phi(G)$ and Lagrange inversion for their coefficients; singularity analysis (branch points, poles, Puiseux expansions) to extract the radius of convergence and local behaviour; and the *transfer theorem* to convert local singularity data into asymptotic coefficient growth via (6). These results are standard but rarely invoked in stochastic-field-theory practice; the classification of universality classes by a rational invariant (p/q) is foreign to the case-by-case RG tradition.

The recognition is the bridge. For a reader who knows Doi–Peliti well (generator construction (2), propagator, vertices, Feynman diagrams), the Lagrange-equation form (5) is the same object written in different notation. What is *not* usually known is that the Flajolet–Sedgewick theorem apparatus transfers wholesale: the transfer theorem, Puiseux classification, and branch-gap instanton integral are all available without any further derivation. Theorem 1 makes the transfer explicit.

5 What can be read off from any CRN

For a CRN with polynomial vertex structure, the CFAC pipeline (Corollary 2) produces, from the stoichiometry matrix alone:

- The Liouvillian Q in closed form (Eqs. (2)–(1)).
- The vertex dictionary $\{g_{mn}\}$ by expansion.

- The vertex function $\phi(G)$ by Eq. (4).
- The Lagrange equation $G = z\phi(G)$ by substitution.
- The branch point (G_*, z_*) by solving $1 - z_*\phi'(G_*) = 0$ together with $G_* = z_*\phi(G_*)$.
- The Puiseux exponent p/q by the Newton polygon of (5) at the branch point.
- The all-orders perturbative coefficient asymptotics by Eq. (6).
- The universality class via p/q (mean-field).
- The nucleation barrier S for bistable systems by the branch-gap integral (7).
- The upper critical dimension d_c from the engineering dimensions of the vertices.
- The UV divergence structure at any loop order by power counting on the finite 1PI graph enumeration.
- **The one-loop bridge integrals** in closed algebraic form (Corollary 5): each one-loop two-propagator integral equals $(1/\sqrt{\lambda_E})\ln[(\sqrt{\lambda_E}+p^2+m_1^2+m_2^2)^2/(4m_1^2m_2^2)]$ with λ_E the Euclidean Källén discriminant of the graph—no numerical quadrature, branch cuts read off the Newton polytope.

At higher loops the bridge integrals are no longer closed-form symbolic but are tabulated once (Bogner and Weinzierl, 2010) and evaluated exactly by tropical Monte Carlo (Borinsky, 2020; Borinsky et al., 2025), reused across CRNs. When the ω-enumeration of the final bullet establishes one-loop UV-exactness—the standard situation for reaction networks of physical interest—the whole pipeline is therefore closed-form symbolic end-to-end: no quadrature, no table lookup, no Monte Carlo.

6 Worked example: pair annihilation

Single-species pair annihilation $2A \to \varnothing$ with rate λ has stoichiometry $k_A = 2$, $l_A = 0$. The Liouvillian (2) is $Q = \lambda[1-(1+\tilde{\psi})^2]\psi^2 = -2\lambda\,\tilde{\psi}\psi^2 - \lambda\,\tilde{\psi}^2\psi^2$. Vertex dictionary: $g_{12} = -2\lambda$, $g_{22} = \lambda$. Vertex function by (4):

$$\phi(G) \;=\; 1 + g_{12}\,G + g_{22}\,G^2 \;=\; 1 - 2\lambda\,G + \lambda\,G^2. \tag{10}$$

Lagrange equation (5): $G(z) = z\big[1 - 2\lambda G + \lambda G^2\big]$, a quadratic in G. Branch point: $(G_*, z_*) = (1/\lambda, 1/\lambda)$; Puiseux exponent $p/q = 1/2$. Transfer theorem (6): $[z^n]G(z) \sim n^{-3/2}\lambda^n$.

Weyl's law on an arbitrary graph with spectral dimension d_s converts this to the density decay $\rho(t) \sim t^{-d_s/2}$ for $d_s \geq 2$, with its well-known mean-field breakdown for $d_s < 2$ (Täuber, 2014). No loop integral was evaluated at any point. A fully worked example for the Gribov/BRW process is given in a companion note (Amarteifio, 2026a).

7 Coupled particle-field extension

For coupled systems (Doi–Peliti particles + Martin–Siggia–Rose field with interaction vertices), the DSE becomes a *system* of Lagrange equations $G_i = z_i\,\phi_i(\{G_j\})$. Eliminating auxiliary variables by resultant gives a univariate polynomial to which Corollary 4 applies. The perturbative consequences (C1), (C2), (C4) of Theorem 1 extend unconditionally to the coupled setting; the non-perturbative nucleation consequence (C3) extends only while the nucleation saddle remains a real branch point of the resultant. A detailed development—including the demonstrations of one-loop perturbative exactness, a branch-gap instanton formula verified to six digits on seven parameter sets in the well-mixed limit, and a three-loop two-colour Kirchhoff factorisation, together with a 3-site lattice test that identifies the failure mode of (C3) at moderate spatial coupling—appears in (Amarteifio, 2026b).

8 Discussion

The content of Theorem 1 is structural. It does not shorten any *single* Doi–Peliti calculation—the construction of Q, the extraction of vertices, the writing down of the DSE are all standard. What it changes is what you can *read off* once you have $\phi(G)$: the transfer theorem gives all-orders asymptotics that the usual iterative loop expansion does not produce; the Puiseux exponent classifies mean-field universality as a rational invariant rather than a case-by-case RG result; the branch-gap contour integral gives non-perturbative nucleation rates from the same polynomial that controls the perturbative series. The apparatus was always available in analytic combinatorics; what was missing in the stochastic-field-theory literature was the recognition that it applies.

Two structural consequences deserve emphasis. First, the coincidence of the branch point in (C1) and (C3) implies that perturbative and non-perturbative physics sit at the same singularity of the same generating function—a connection obscured in the direct-RG route, which treats loop corrections and instantons with distinct technology. Second, universality classification by Puiseux exponent p/q is a statement about the polynomial ϕ alone; two CRNs with the same ϕ share a universality class regardless of the reactions that produced them. This makes the mapping from biology to physics (which CRN belongs to which class) a question of polynomial equivalence.

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